SQL COMMIT AND ROLLBACK:

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🏢 Lesson: COMMIT and ROLLBACK in SQL (MySQL)

🔷 1. Introduction:

👉 **In databases, sometimes we run multiple queries that depend on each other.**

For example:

- If we are transferring money from Alice to Bob, we must **deduct from Alice** AND **add to Bob**.
- If one query succeeds and the other fails, the database will become **inconsistent**.

Transactions help us solve this problem.

- **COMMIT** → Save all changes permanently.
- **ROLLBACK** → Undo all changes since the transaction started.

🔷 2. Create a Table

```
CREATE TABLE accounts (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   balance DECIMAL(10,2)
)
```

🔷 3. Insert Initial Data

```
INSERT INTO accounts (name, balance) VALUES
('Alice', 5000.00),
('Bob', 3000.00);
```

Check the data:

```
SELECT * FROM accounts;
```

id	name	balance
1	Alice	5000.00
2	Bob	3000.00

🔷 4. Start a Transaction

```
START TRANSACTION;
```

👉 This means:

- From now on, MySQL won't save changes permanently until we say **COMMIT**.
- We can still undo everything using **ROLLBACK**.

◈ 5. Perform Updates (Money Transfer)

Deduct from Alice:

```
UPDATE accounts SET balance = balance - 500 WHERE name = 'Alice';
```

Add to Bob:

```
UPDATE accounts SET balance = balance + 500 WHERE name = 'Bob';
```

Now check:

```
SELECT * FROM accounts;
```

id	name	balance
1	Alice	4500.00
2	Bob	3500.00

⚠ **Important:** These changes are visible in your session, but not yet permanent.

◈ 6a. COMMIT Example (Save Permanently):

```
COMMIT;
```

☞ Now changes are written permanently to the database.

- Even if you disconnect MySQL, the changes stay saved.

◈ 6b. ROLLBACK Example (Undo Changes):

Instead of COMMIT, let's try again:

```
START TRANSACTION;
UPDATE accounts SET balance = balance - 500 WHERE name = 'Alice';
UPDATE accounts SET balance = balance + 500 WHERE name = 'Bob';
ROLLBACK;
```

Check again:

```
SELECT * FROM accounts;
```

id	name	balance
1	Alice	5000.00
2	Bob	3000.00

☞ The changes are **undone**, database is back to original state.

◈ 7. SAVEPOINT Example (Partial Rollback)

Sometimes we want to undo only part of a transaction.

```
START TRANSACTION;
-- First transfer
UPDATE accounts SET balance = balance - 500 WHERE name = 'Alice';
```

```
UPDATE accounts SET balance = balance + 500 WHERE name = 'Bob';
SAVEPOINT after_first_transfer;
-- Second transfer
UPDATE accounts SET balance = balance - 200 WHERE name = 'Alice';
UPDATE accounts SET balance = balance + 200 WHERE name = 'Bob';
-- Oops! Something wrong with second transfer
ROLLBACK TO after_first_transfer;
COMMIT;
```

👉 Only the **first transfer** happens, the second one is undone.

🎯 **Final :**

- **COMMIT = Save** (like pressing *Save* in MS Word).
- **ROLLBACK = Undo** (like pressing *Cancel*).
- **SAVEPOINT = Draft Save** (undo only the latest part, not everything).